Assembly of the precise inverse-theorem input

Author successor: /root/astra_v14_baseline_theta

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Status. This successor assembles the exact local proofs identified below and retains their separate independent decisions. The preceding assemblies' local splitting, selection, analytic and finite-family joins have conditional acceptance. The corrected base, degree conversions, rank iteration and final Section 12 and Section 13 units have their own scoped decisions. The local converse and plain additive-quadruple argument have also been independently accepted. The actual Sections 9–11 reconstruction has conditional acceptance. The one-variable Appendix D/F packet has bounded decisions. This successor connects its affine-offset transfer and the multivariable E.11/E.13 uses to the independently reviewed local multifactor supplier. Its upstream and downstream decisions are separate. The new composition and remaining source-coverage review must still be assessed; no release approval is presumed here. This document does not convert scoped decisions into unconditional GI or whole-source acceptance. Exact versions, decisions and open interfaces are bound in the accompanying records.

1 Exact target and conventions

The target, denoted GI(k), is the following unchanged linear formulation. For each fixed integer $k \geq 3$ and each $0 < \delta \leq 1$, there are a finite family $\mathcal{M}_{k,\delta}$ of nilmanifolds G/Γ, each with connected simply connected nilpotent group of step at most k and a discrete cocompact lattice, fixed smooth metrics, and constants $C, c > 0$, such that every 1-bounded $f : [N] \to \mathbb{C}$ with $\|f\|_{U^{k+1}[N]} \geq \delta$ correlates with a 1-bounded test F of Lipschitz constant at most C along a linear orbit:

$$\left|\mathbb{E}_{n\in[N]} f(n)F(g^n x)\right| \geq c.$$

The family and constants depend on k, δ, not on N, f or the orbit parameters. Strict source thresholds can be used with $\delta/2$ to give this weak-inequality version. No effective numerical bounds are asserted.

For $d \geq 2$, the interval norm is the norm of the zero extension to a cyclic group of sufficiently large order divided by the norm of the interval indicator, as in the published GTZ definition on p. 1232. We use the same normalization at every inverse step. On a finite abelian group A, write

$$\|v\|_{U^d(A)}^{2^d} = \mathbb{E}_{x,t_1,\dots,t_d\in A} \prod_{\omega\in\{0,1\}^d} \mathcal{C}^{|\omega|} v(x + \omega \cdot t),$$

where $\mathcal{C}$ is complex conjugation. Our correlation convention is $|\mathbb{E} f\psi|$; conjugating the test function converts the other common convention without changing its degree or bounds.

All limit nilcharacters in the assembly are internal in the explicitly specified sense of the local Lemma 13.2 and tautification units: the vertical identity holds for all limit group elements and all limit points, with the natural extension of a standard vertical character. Every constructor used

below must supply that domain, rather than only standard-point equivariance. This is an explicit application check, not a consequence of bounded Lipschitz continuity alone.

The manuscript currently requires GI(20) and GI(39). The proposed relative inverse replacement also requires GI($r-1$) for each fixed $r = \max(s, 4)$ chosen before the norm threshold and asymptotic variable. A polynomial orbit without a same-step linearization or an N-dependent family does not satisfy this target.

2 Intermediate interfaces that must be supplied

The following lists the available components and the remaining mathematical premises of the assembly argument. A local decision applies only to its stated scope; the incoming source contracts remain explicit.

1. The compatible bounded polynomial inverse theorem of order 2, proved in `gi-base-002` and independently accepted relative to its retained published forward bracket theorem and Bohr-set inputs. It gives the fixed finite family and normalized interval U^3 norm. The needed trivial-subgroup specialization of the false quadratic extension is proved locally, and the periodic connected polynomial construction replaces the affected skew-iterate route.

2. The nilcharacter approximation/selection interface: from positive correlation with a bounded degree-d nilsequence, obtain a degree-d unit-vector nilcharacter and a scalar component giving positive correlation. For a dense family of parameters, make the choices in a limit fashion and, after passage to a dense subfamily if necessary, keep the finite dimensions, complexities, bounds and positive correlation standard and uniform. Section 4 below gives the finite-net and unit-vector construction used at this interface, with conditional independent acceptance against the previous finite inverse family. Later selection applications retain their own source contracts.

3. Linearization at inverse order s: for a dense family of degree $s-1$ nilcharacters χ_h correlated with $\Delta_h f$, obtain one multidegree $(1, s-1)$ nilcharacter $\chi(h, n)$ with which $\Delta_h f$ has degree-at-most-$s-2$ correlation for many h. This is the precise Theorem 7.1 interface. The modified Proposition 7.3 is supplied by `local-proof-sol-lp02-03-r3`; the quotient and rank-lower comparison by `lp04-001`; the final orbit by the accepted Section 12 successor; and the degree conversions, unit-vector recovery, finite rank iteration and rank-zero endpoint by `rank-iteration-001`. The frozen Sections 9–11 reconstruction supplies an independently reviewed proof of the one-step frequency premise at every newly constructed rank family, conditional on its precise one-variable D and F contracts. Its theorem takes the same uniform unit-vector degree–rank family and accumulated multidegree factor as the rank iteration. It reconstructs all core/linear/independent data anew; none is inherited. Its fixed unit padding is removed by projecting the first factor of the entire LP04 matrix, so the matrix is exactly $\chi_h \otimes \overline{\zeta_h}$ required by the next rank step. The Section 12 successor supplies the same fixed coordinate isometry as the earlier version named in the frozen rank packet; its only declared change makes the zero-index groups explicit. The new binding is recorded here without rewriting that earlier freeze. All these steps retain the full-limit vertical identities and uniform bounds.

   For clarity, the invariant concerns represented families: ordinary unit-vector tests with exact global vertical identities, fixed standard bounds and finitely many standard models and characters. The initial positive-degree selection is made among these representations, by the

parameter-level selection argument for the preceding inverse order; it does not upgrade an arbitrary weak nilcharacter or preserve a prescribed symbol. Degree-to-rank conversion retains the represented function and restricts its character, so this invariant starts at (m, m), where $m = s - 1$, with accumulated factor $A = 1$.

Distinguish the fixed inclusion J that pads the incoming χ_h in Sections 9–11 from the final Section 12 inclusion I, for which $Z(h, n) = I\zeta_h(n)$ on the retained shift set. The full ordinary-top character in Section 12 is chosen before the LP04 comparison and restricts to its cancelling rank-top character. The comparison thus supplies the entire vector

$$W_h = (J\chi_h) \otimes \overline{\zeta_h}$$

on one common model of degree-rank at most $(m, R - 1)$. If $QJ = \mathrm{id}$ is the coordinate projection on the first factor, then

$$(Q \otimes \mathrm{id})W_h = \chi_h \otimes \overline{\zeta_h} =: V_h.$$

This fixed bounded linear map preserves the common model, internal parameter dependence and standard Lipschitz bound. The projection recovering ζ_h from Z is instead the one associated with I. In particular, this conclusion concerns every matrix entry, not a separately selected scalar entry.

The identity $\chi_{h,a} = \sum_b \zeta_{h,b}(\chi_{h,a}\overline{\zeta_{h,b}})$ uses the unit norm of the whole vector ζ_h and therefore uses exactly V_h in the correlation recovery. The finite ordinary restart dictionary preserves the global unit and vertical identities after internal selection; the ordinary tests chosen in the proof of E.5 are extended only after their finite selection. Tensoring the accumulated factor with Z preserves its unit norm and multidegree $(1, m)$. Sections 9–11 then reconstruct all frequency data for this new family. No ordinary degree drop is claimed before the rank endpoint: at rank zero the degree-rank axiom makes the remaining factor ordinary degree at most $m - 1$, which is then absorbed into the lower-degree test. The geometric, analytic and source premises of all these component constructions remain those of their stated contracts.

The centered fractional-part convention in this chain is $(-1/2, 1/2]$, a fixed fundamental cube permitted by Section 12. The reconstruction's acceptance remains conditional on its suppliers; the affine-offset coefficient transfer is supplied by the local multifactor lemma, as connected below.

4. Symmetrization at inverse order s: from that input obtain one degree-s nilcharacter Θ and a bounded nilsequence $\Psi(h, n)$ supported on

$$J = \{(i, j) : i + j \le s - 1\} \ \cup \ \{(i, s - i) : 2 \le i \le s\}, \tag{1}$$

such that the relevant $\chi(h, n)$ is a bounded finite linear combination of components of $\Theta(n + h) \otimes \overline{\Theta(n)} \otimes \Psi(h, n)$. This is the required Theorem 7.4 output, supplied by the locally authored Section 13 proof with its scoped independent decision. It takes the degree-at-most-$s - 2$ correlation actually supplied above; the printed strict hypothesis is not used. Its proof binds the corrected multilinearization, literal block symmetry, local Lemma 13.2 and later shift/sign calculations. It also retains the precise stronger Proposition 8.4, E.11 and E.13 applications. The precise plain Proposition 8.4 input is supplied by the accepted local `plain-quadruples-001`: select the same scalar coordinate of $\chi(h, \cdot) \otimes \phi_h$ and retain that entire lower factor. For $H \subset [-N, N]$ of density α, correlation δ and global scalar bound B, that proof gives standard positive density and correlation depending only on α, δ, B. Its output

has signs $++--$ and shifts in the second and fourth factors by $h_1 - h_4$, exactly as used in Section 13. The functions have global polynomial representations; the interval cutoff belongs to f. The proof removes its remaining bounded cutoff by an inequality, not by an asserted identity. Finite component selection and the internal version preserve standard uniform constants. This is an explicit substitution of the supplying proof for the published Proposition 8.4 invocation in the frozen Section 13 proof, lines 88–125. That earlier file remains unchanged and does not itself refer to the later local proof. Its intervals $I_1 = [-N, N]$ and $J = [N]$, scalar component, global test domain, and quadruple count on the N^3 scale are exactly the local theorem's data. The reparametrization $(h_1, h_2, h_3, h_4) = (h_0 + a, h_0 + b, h_0 + a + b, h_0)$ is therefore unchanged; the subsequent pigeonhole has $h_0 \in I_1$ and $a, b \in [-2N, 2N]$. The E.11 and E.13 inputs require the actual multivariable Appendix D.5 factorization and all-box reconstruction; a one-variable D.4 proof does not supply that premise. This obligation is supplied by the local multifactor theorem and its two independent decisions, as connected below.

5. Splitting for a union of two finite downsets: a fixed bounded nilsequence on their union is uniformly approximable, to any fixed positive error, by a finite sum of products of nilsequences supported on the two downsets, with fixed finite complexity and bounds. This is the E.4 application used explicitly in the next section.

6. The converse correlation bound: a bounded scalar function with positive correlation against a bounded degree-d nilsequence has positive $U^{d+1}[N]$ norm, with a bound for each fixed complexity and correlation. The local `converse-001` proves the required implication using the published compact cube constraint, an explicitly normalized smooth average, a uniform Fourier bound and Gowers–Cauchy–Schwarz. It uses accepted exact same-step linearization for polynomial tests. It does not import the disputed averaging-domain transition in Green–Tao 11.2 or invoke the inverse theorem being proved. Its exact source application and independent decision are bound separately.

7. Countability, up to filtered isomorphism, of the fixed rational filtered nilmanifolds of bounded degree, and equivalence of fixed smooth metrics on a compact nilmanifold. The published countability assertion is on p. 1250 with its Mal'cev-coordinate justification. Its exact scope must include the taut rational objects constructed by this project.

8. The local increment linearization theorem: an exact polynomial nilsequence of degree s has a same-step linear representation on fixed geometry with a fixed Lipschitz comparison. This is supplied by the separately authored and independently accepted $P_1 \rtimes \mathbb{R}$ unit, rather than by the defective full-P_0 suspension in Appendix C.

The increment unit supplies the exact sequence and finite-family join used here. It does not assert the stronger clause in the original Proposition C.2 about one universal projection into the original, unnormalized nilmanifold for every polynomial: bounded normalization is absorbed in the test function F_q. That distinction changes no part of the target in section 1.

The successor connections to the frozen multifactor proof are as follows. They select a supplying proof for the existing conclusion and preserve the old component files. Their independent decisions are recorded separately when available.

First, the one-variable weak factorization uses a progression $A+b[M]$ with $|A| = O(N)$, standard bounded nonzero integer b, and $M \asymp N$. Apply the local inverse-affine lemma with one variable to a real lift of the horizontal polynomial. Its input is the positive binomial coefficient bound on

$m \mapsto f(A + bm)$. In the inverse substitution, the coefficient from input degree j to positive output degree i has size $O(N^{j-i})$ and a standard denominator independent of A. One standard integer multiplier clears the integer portions; the residual $O(N^{-j})$ becomes $O(N^{-i})$. Thus the same original horizontal character, multiplied by a standard positive integer, has the required ambient coefficient bound. It remains nonzero. No bound on the constant coefficient is asserted or needed. This is the exact proof-level transfer at DF002 lines 132–145, distinct from its right-coset periodicity argument.

Second, at Section 13 lines 327–343 take the local multifactor bias theorem with dimension $s + 1$, degree s, and its scale equal to $8N$. The input is the actual reflected unit nilcharacter σ_ε and reflected convex polytope obtained by the orthant selection. Every positive progression is the full $[8N]$; the auxiliary lower-degree test is constant. Its conclusion supplies a globally represented lower-degree nilsequence agreeing with σ_ε on all of $[-8N, 8N]^{s+1}$. The inverse reflection is an integer linear substitution, so the same full-box conclusion holds for σ. This does not extend an identity from only the selected positive orthant by assumption.

Third, after the displayed diagonal pullback in Section 13, the actual unit tensor-product representative $\xi(h, n)$ of the antisymmetric symbol satisfies $s![\xi] = 0$ on $[-8N, 8N]^2$. By the definition of the zero symbol, the coordinates of $\xi^{\otimes s!}$ are lower-degree nilsequences on that box; their finite product of models is one common lower-degree model. Apply the local torsion theorem in dimension two, degree s, scale $8N$, and positive standard tensor power $s!$. It gives a globally represented degree-at-most-$(s-1)$ nilsequence equal to ξ on that same symmetric box. Only afterward restrict to $[-N, N] \times [N]$, as in Section 13 lines 345–363.

There is also a distinct converse use at Section 13 lines 298–305. For $s \geq 4$, put $d = s - 3 \geq 1$ in the accepted local converse and apply it to the uniformly selected lower-degree test correlated with the chosen coordinate of α'. This gives the positive $U^{s-2}[N]$ bound there, uniformly after the stated selections. For $s = 3$, a degree-zero nilsequence is constant: a fixed positive correlation with a bounded constant directly implies a fixed positive $|\mathbb{E}_{n\in[N]}\alpha'(n)|$: if the correlation is at least $c > 0$ and the constant has modulus at most M, this mean has modulus at least c/M. Zero extension has group mean $(N/|Q|)\mathbb{E}_{[N]}\alpha'$, and the interval indicator has U^1 quantity $N/|Q|$, so normalization gives exactly that interval mean. No degree-one test is inserted into this case.

Closure of fixed-degree nilsequences under conjugation, scalar multiplication and finite products uses the actual product nilmanifold and product filtration. The top degree is the maximum of the input degree bounds. This elementary object identification is also required when it is used in the induction below.

3 Correct splitting of the symmetrization remainder

We first give the lattice detail needed by the splitting application. It uses the integral Hall-coordinate lemma in the independently reviewed LP04 unit, not a general assertion that disjoint rational Lie subgroups automatically separate points after quotienting by a lattice.

Lemma 1 (Splitting on two finite multidegree downsets)**.** *Let $J_1, J_2 \subset \mathbb{N}^a$ be finite downsets. Every bounded Lipschitz nilsequence of multidegree supported on $J_1 \cup J_2$ is uniformly approximable, to each fixed positive error, by a finite sum of products of nilsequences supported on J_1 and J_2. The construction uses fixed geometries and bounds before the polynomial coefficients or the interval length vary, for each fixed input geometry and test bound.*

Proof. Use bounded tautification to set the polynomial's value at zero to the identity and restrict the ambient group to the group generated by its positive coordinate groups. For each positive $j \in J_1 \cup J_2$ choose a finite generating set of $\Gamma \cap G_j$. Its real generated group is G_j, by the rational

subgroup assertion. Introduce one formal letter of weight j for each such generator. Take the free real Lie algebra on these letters and discard every individual-letter multidegree whose total vector weight is outside $J_1 \cup J_2$. This is an ideal: the complement of a downset is upward closed. Every letter has positive total weight, so the surviving length is bounded by $\max_{j \in J_1 \cup J_2} |j|$.

Let U be the exponential group and Λ the group generated by the exponential letters. The Hall-coordinate lemma proves that Λ is a lattice and that all weight filtration groups are rational. Sending each letter to its chosen original lattice generator gives a filtered Lie homomorphism $\phi : U \to G$ with $\phi(\Lambda) \subseteq \Gamma$. It maps U_j onto G_j: the letters of weight exactly j already generate G_j as a real group. Lift each Taylor coefficient of the normalized original polynomial through this surjection. The multivariate Taylor theorem, B.9, makes their ordered binomial product an adapted polynomial $\widetilde{g}$ with $\phi\widetilde{g} = g$. Pulling the test back along $U/\Lambda \to G/\Gamma$ therefore preserves the sequence exactly. Its Lipschitz bound increases by a fixed geometric factor only.

For $\nu = 1, 2$ let I_ν be the connected ideal subgroup spanned by the surviving Hall words whose weight is outside J_ν. The quotient U/I_ν has degree support in J_ν. The integral Hall lemma says that its quotient lattice is obtained by deleting precisely those coordinates. No surviving Hall word is deleted by both quotients, since its weight belongs to $J_1 \cup J_2$. Consequently the map

$$U/\Lambda \longrightarrow U/(I_1\Lambda) \ \times \ U/(I_2\Lambda)$$

is injective. In detail, if two cosets have equal images, their difference has integral coordinates in each projected Hall system. The union of the retained coordinate sets is the full Hall system, so every coordinate of the difference is integral. The difference is in Λ. This checks the lattice condition as well as the Lie algebra condition.

The pullbacks of the two algebras of Lipschitz functions on these compact quotients contain the constants, are closed under conjugation, and together separate points by that injection. The complex Stone–Weierstrass theorem approximates the pulled-back test uniformly by the algebra they generate. Each element of that algebra is a finite sum of products with one factor from each quotient, since each of the two individual algebras is closed under multiplication. Composing with $\widetilde{g}$ gives the asserted splitting. For an internal bounded Lipschitz test, first approximate by one standard Lipschitz test on the fixed compact manifold, using Definition 5.1, and then perform this standard finite approximation. For a fixed uniform Lipschitz and norm bound, compactness of that class in the uniform norm supplies a finite net and thus fixed finite bounds. Only this fixed-bound form is used below. If the positive groups are trivial, the sequence is constant and the assertion follows with one constant product. □

The coordinate order is (h, n). Set

$$A = \{(0, j) : 0 \le j \le s - 1\}, \qquad B = \{(i, j) : j \le s - 2, \ i + j \le s\}.$$

Both are downsets and $A \cup B = J$. Indeed the only point of J with $j = s - 1$ is $(0, s - 1)$, covered by A. Every other point of J has $j \le s - 2$ and lies in B. Conversely a point of B of total degree s has $i \ge 2$, so belongs to the second part of (1); all its other points lie in the first part. This proves equality in both directions.

Apply interface 5 to the actual Ψ with these two downsets. A sequence supported on A is independent of h: its polynomial map has every h-direction derivative in the trivial first-coordinate group. It is a degree-at-most-$s - 1$ function $\psi(n)$ of n alone. For a factor supported on B, fixing h gives a polynomial nilsequence in n of degree at most $s - 2$, since all higher n-direction groups are trivial. If normalization of this section at fixed h is needed, the accepted bounded tautification theorem supplies it with uniform geometry and parameter dependence.

Here is the needed approximation-to-correlation connection. After uniform selection, suppose the preceding correlation is at least a standard $\epsilon_0 > 0$ on a dense set of h, and the lower-degree test factors have magnitude at most a standard M. Choose the uniform splitting error less than $\epsilon_0/(2M)$, also absorbing the fixed finite component and coefficient bounds into M. The error in each normalized correlation is then at most $\epsilon_0/2$. A finite sum remains, so one summand has a positive standard correlation on a dense subfamily by finite pigeonhole. The same finite choice of components of Θ can be made on a dense subfamily. The result is two fixed bounded scalar nilsequences θ, θ' of degree at most s such that

$$n \longmapsto f(n+h)\overline{f(n)}\theta(n+h)\theta'(n) \tag{2}$$

is degree-at-most-$s-2$ biased for a dense set of h. The B-factor is absorbed into that lower-degree test. The A-factor $\psi(n)$ is multiplied into the unshifted component forming θ'. No scalar component is divided by, or assumed to have modulus one.

The printed source on p. 1272 uses the opposite coordinate restrictions while claiming independence of h. The corrected downsets and the approximation estimate above are the explicit replacement for that step; its historical printed form is preserved in the findings record.

4 Uniform nilcharacter selection and fixed geometries

Here is the selection needed at the start of each induction step. Use the finite-family inverse statement already obtained at the previous order; the finite-family extraction below is performed after each order, before proceeding to the next. This avoids assuming that an arbitrary external assignment of nilmanifolds to parameters is an internal family.

On one fixed taut degree-d nilmanifold, the top group is central and its quotient by its lattice is a compact torus T. For each integral character η of T, a unit-vector smooth function with that vertical character exists as follows. Cover the compact quotient by T by finitely many trivializing charts. Choose smooth real functions ρ_j supported inside the charts with $\sum_j \rho_j^2 = 1$, using a smooth partition of unity and normalizing its vector by its positive Euclidean norm. In the jth trivialization define the jth component by $\rho_j(b)e(\eta(t))$, and extend it by zero off that chart. These components are smooth, have the exact character, and their squared magnitudes sum to one. The central rational quotient and its lattice are justified by the accepted quotient geometry in the local Lemma 13.2 unit. This is the explicit construction used in published Lemma 6.4.

Components of such unit-vector functions, including frequency zero, form a conjugation-stable multiplicative family: use tensor products for multiplication and conjugation for conjugation. They contain constants and separate points. Distinct points on the same T-fibre are separated by an integral character, with a nonzero component of the unit vector. Points on different fibres are separated by a smooth frequency-zero scalar unit function pulled back from the base. Stone–Weierstrass therefore gives uniform approximation of every continuous test by a finite linear combination of these components.

For the fixed norm and Lipschitz bound in the previous inverse theorem, Arzela–Ascoli gives a finite uniform net of tests. Apply the preceding approximation to each element of that net with error smaller than a fixed fraction of the positive inverse correlation. There are finitely many models and net elements, so the total list of unit-vector test maps, dimensions, frequencies, coefficient bounds and Lipschitz bounds is fixed. Pigeonholing a scalar component of an approximating sum preserves a fixed positive correlation. No component is assumed to have modulus one; the entire vector has norm one.

For each finite interval and each of its good shifts, ordinary choice selects one polynomial orbit and one item from this fixed finite list. Taking the ultralimit of these finite assignments gives an internal family. A finite pigeonhole step restricts to a dense set of shifts with one common fixed model and fixed test map if required. The unit and vertical identities hold at every representative and hence for all limit arguments, with the natural extension of the chosen standard character. This establishes the selection interface needed here from the preceding finite inverse theorem. It does not replace separate selection applications later in Sections 8–13, which remain with their actual consumers.

For the countability interface, use the rational Mal'cev-coordinate description of a lattice in a connected simply connected nilpotent Lie group, as invoked in published Section 3 and p. 1250. In each finite dimension the rational structure constants have countably many possibilities. The rational subspaces defining the finitely many filtration groups are specified by finite matrices with rational entries, again a countable choice. In the rational structure determined by the lattice, a finite list of rational logarithmic generators specifies the lattice. Thus there are at most countably many such filtered lattice data up to isomorphism. All local constructions here have finite rational filtrations and discrete cocompact lattices, so belong to this class. On each compact quotient any two fixed smooth Riemannian metrics are bi-Lipschitz equivalent: compare their positive quadratic forms on the compact unit tangent bundle. These are the exact countability and metric comparisons used in the extraction below.

5 The finite-group inequality needed for the last induction step

Lemma 2. *For bounded complex functions a, b on a finite abelian group Q and an integer $d \geq 1$,*

$$\mathbb{E}_{h\in Q}\|a(\cdot+h)b(\cdot)\|_{U^d(Q)}^{2^d} \leq \|a\|_{U^{d+1}(Q)}^{2^d}\|b\|_{U^{d+1}(Q)}^{2^d}.$$

Proof. For $t = (t_1, \dots, t_d)$ write $C_a(x;t) = \prod_\omega \mathcal{C}^{|\omega|}a(x+\omega\cdot t)$, and define C_b likewise. Expanding the left side and using the bijection $(x,h) \mapsto (x, y = x+h)$ gives

$$\mathbb{E}_t(\mathbb{E}_y C_a(y;t))(\mathbb{E}_x C_b(x;t)).$$

Its absolute value is at most $\left(\mathbb{E}_t|\mathbb{E}_y C_a(y;t)|^2\right)^{1/2}\left(\mathbb{E}_t|\mathbb{E}_x C_b(x;t)|^2\right)^{1/2}$. Each squared expression is the $(d+1)$-cube average: expand the square and substitute its second base point as the first plus a new increment. It therefore equals the corresponding U^{d+1} norm to the power 2^{d+1}. This proves the inequality. The original left side is nonnegative because it is an average of powers of Gowers seminorms. □

Choose a cyclic group Q with size between C_sN and $2C_sN$ for a fixed sufficiently large C_s, and zero-extend all functions supported on $[N]$. The normalized interval and group norms differ by the fixed indicator factor. For each fixed order that factor is bounded above by one and below by $N/|Q|$, using the finite-group Gowers–Cauchy–Schwarz inequality. Thus a fixed positive interval norm gives a fixed positive group norm, and conversely; the constants depend only on the fixed ambient ratio and order. Taking C_s larger than 2^{s+1} ensures the same ambient group is valid for all interval orders used here. The zero-extended product in (2) is exactly the group product for $|h| \leq N$; there is no wrapped overlap since the ambient size is larger than $4N$.

By interface 6, (2) has a uniformly positive $U^{s-1}[N]$ norm for a dense set of h. Let $a = f\theta$ and $b = \overline{f}\theta'$, both zero-extended. The average over all group shifts of $\|a(\cdot+h)b\|_{U^{s-1}(Q)}^{2^{s-1}}$ is bounded below by a fixed positive constant: the contributing set of interval shifts has positive density also in

Q, and the other summands are nonnegative. Lemma 2 and the fixed bound on b imply $\|a\|_{U^s(Q)}$ is positive by a standard amount, hence so is $\|f\theta\|_{U^s[N]}$. The inverse theorem at order $s-1$ now makes $f\theta$ correlate with a degree-at-most-$s-1$ nilsequence ψ. The actual product $\theta\psi$ is a degree-at-most-s nilsequence on the product filtered nilmanifold. Thus f is s-biased.

6 Starting and completing the induction

The bounded induction base is the separately frozen and reviewed `gi-base-002`. It starts from the exact forward bracket theorem, proves the affected trivial-subgroup specialization, constructs a periodic connected polynomial test and transfers its correlation directly to the normalized interval. The older base in assembly001 is superseded and remains historical. No unpublished correction is treated as an accepted black box.

For completeness the initial derivative step has the same normalization control. The cube identity on the fixed ambient cyclic group gives

$$\mathbb{E}_h \|\Delta_h f\|_{U^s(Q)}^{2^s} = \|f\|_{U^{s+1}(Q)}^{2^{s+1}}.$$

If the interval norm is positive by a standard amount, so is the right side. Since f is bounded, all summands have a fixed upper bound. The nonzero differences occur only on the shifts represented by $[-N, N]$, so a dense subset of those shifts has a uniformly positive $U^s[N]$ norm. Apply the inverse theorem of order $s-1$, then interface 2, to obtain the family required by interface 3. Interfaces 3–5 and the preceding section complete the induction step. The bounded limit-N case is included: a bounded interval has standard finite size, and any function on it is represented by a Lipschitz function along a rational circle rotation, with a standard finite Lipschitz bound. Thus the induction proves the limit polynomial inverse conclusion for every limit N, conditional on the listed intermediate interfaces and base.

We spell out the finite-family extraction rather than replace it by pointwise compactness. Enumerate the rational filtered nilmanifolds of degree at most s up to isomorphism as $M_1, M_2, \ldots$, with one fixed smooth metric on each, using interface 7. If the finite polynomial inverse statement failed at some fixed $\delta > 0$, for each j choose a 1-bounded $f_j : [N_j] \to \mathbb{C}$ with norm at least δ, but correlation at most $1/j$ against every polynomial nilsequence from the first j manifolds with 1-bounded test and Lipschitz norm at most j. Its ultralimit has positive norm, so the limit inverse conclusion gives a fixed standard filtered nilmanifold, bounded Lipschitz limit test and positive standard correlation. Normalize the test by a fixed bound to make it 1-bounded. It is isomorphic to some fixed M_{j_0}; fixed metric equivalence transfers its Lipschitz bound to a standard constant. Representatives of the polynomial and test satisfy the same bounds with one slightly larger standard constant on an ultrafilter-large set. For j larger than that constant and j_0, the selected correlation upper bound tends to zero, contradicting the positive lower bound. This yields the finite-family polynomial statement with all parameters fixed independently of N.

Finally interface 8 converts every member of the finite polynomial family to a fixed nilmanifold of step at most s, with a fixed Lipschitz bound and exactly equal sequence. A finite maximum of those bounds and the unchanged positive correlation give the exact linear target in section 1. This final join depends on the reviewed increment construction; it does not import the failed full-polynomial suspension claim.

7 What remains before this is a complete proof

The local units and their exact incoming connections are now assembled relative to their retained published inputs. The affine coefficient transfer and multivariable factorization, bias and torsion conclusions have separate independent decisions. Their substitutions and the degree-zero converse boundary are new explicit connections in this successor. The earlier composition decision is not silently extended to these new bytes.

The retained published forward bracket/Bohr, quantitative equal-side equidistribution, rational-coordinate and compact cube-constraint inputs keep their exact source scope. Related-use coverage and the complete affected-dependency audit remain separate. This packet is submitted for independent acceptance of the assembled target and its actual current consumer interfaces. It makes no release-incorporation or whole-source approval claim. The target keeps its original normalized norm, quantifier order, fixed finite family and same-step linear output.